

Subject Questions

E-Commerce

IT430

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Question No: 17 __ (Marks: 1)

What is the function of UDP protocol at TCP/IP stack? Give precise answer.

The **User Datagram Protocol (UDP)** is one of the core members of the Internet Protocol Suite, the set of network protocols used for the Internet. With UDP, computer applications can send messages, in this case referred to as datagram's, to other hosts on an Internet Protocol (IP) network without requiring prior communications to set up special transmission channels or data paths.

UDP is a simpler message-based connectionless protocol. Connectionless protocols do not set up a dedicated end-to-end connection.

Question No: 18 (Marks: 1)

What tag is used for displaying a border between the frames?

BORDER or FRAMEBORDER – Set this attribute to 0 for not displaying a border between the frames

Question No: 19 (Marks: 2)

Suppose there is an applet at "http://www.vu.edu.pk/demo.class". Write the code to

Include this applet in your web page.

```
<HTML> <BODY>
<APPLET CODE= "http://www.vu.edu.pk/demo.class". Width=100 height=200>
DEMO </APPLET>
</BODY> </HTML>
```

Question No: 20 __ (Marks: 3)

Identify the object, property and method in the following:

Math.sin(45)

Math.PI

Answer:

Math is the object in above case

PI is the property of Math object

Sin () is the method of Math object

Question No: 21 __ (Marks: 5)

What is a web form? What is its usage in reference to client-server communication?
2+3

A **web form** on a web page allows a user to enter data that is sent to a server for processing. Web forms resemble paper forms because internet users fill out the forms using checkboxes, radio buttons, or text fields. For example, web forms can



be used to enter shipping or credit card data to order a product or can be used to retrieve data

Forms can be combined with various scripting languages to allow developers to create dynamic web sites. This includes both client-side and/or server-side languages.

The de facto standard client-side scripting language for web sites is JavaScript. Utilizing JavaScript on the Document Object Model (DOM) leads to the method of Dynamic HTML that allows dynamic creation and modification of a web page within the browser. While client-side languages used in conjunction with forms are limited, they often can serve to do pre-validation of the form data and/or to prepare the form data to send to a server-side program.

Question No: 22 (Marks: 10)

Explain the following frame tags? Discuss the usage of these

- a) <frameset>
- b) <frame>
- c) <Noframes>
- d) <Iframe>

<frameset>...</frameset>

Contains the frameset The frames layout is given by comma separated lists in the rows and cols attributes.

<frame> or <frame/>

Delimits a single frame, or region, within the frameset A separate document linked with the src attribute appears inside.

<noframes>...</noframes>

Contains normal HTML content for user agents that don't support frames

<iframe>...</iframe>

An inline frame places another HTML document in a frame. Unlike an object element, an inline frame can be the "target" frame for links defined by other elements and it can be selected by the user agent as the focus for printing, viewing its source etc.

Question No: 18 (Marks: 1)

What it stands for IIS?

IIS (Internet Information Server)

Question No: 19 __ (Marks: 2)

When does we use the parseFloat() function?

'Parse Float (string)' parses the string as a floating point number, is the example of a function/method of Global Object.

Question No: 20 (Marks: 3)

Name the JavaScript function that convert strings to float numbers. How do you give the function a string value to convert to a number?

JavaScript provides two functions to convert this string value to a numeric data type; `parseInt()` and `parseFloat()`. By using parse float function we can convert strings to float numbers. To convert string to integer we can use `parseInt` function. Using this function we can only convert string numbers to numeric data but alphabetic characters we can't convert.

Question No: 21 __ (Marks: 5)

Write your personal profile in a file saving it with the extension of e.g. "abc.js" and then call it in a JavaScript code.

Question No: 22 (Marks: 10)

Explain any three of the outside attacks briefly.

Out side attacks on a network

1. Eavesdropping/ sniffing/snooping

In this type of attack the hacker has the ability to monitor network traffic using some kind of network monitoring software. For example, a hacker may install some backdoor or Trojan horse that can monitor the key strokes of a user while typing and send the typed information to the hacker.

2. Password attacks

Such attacks are basically a result of eavesdropping through which the hacker is able to know the account ID or password of a particular user. Then using it the hacker gains access to the network and gather information such as user names, passwords, computer names, resources etc. That can lead to modification, deletion or rerouting of network data.

3. IP address spoofing

You know that there are two IP addresses available on a data packet – IP addresses of the sender and the destination. The address of the destination only matters for routing. It is possible that a hacker (having special capabilities) seizes the control of a router, changes the IP address of the source/sender on data packets and thus forces the destination machine to send the information/web page to a different machine, that is, the machine of the hacker. This is called IP address spoofing.

4. Man in the middle attacks

In it the attacker is able to monitor, capture and control data between sending and receiving machines. He may apply IP address spoofing technique to divert the packets to its machine, then modify the packets and resend the misleading

information to the actual client. Another form of man-in-the-middle attack is where the hacker is able to substitute the IP address of a genuine web site with the IP address of his own web site due to some security hole in the software that runs on a domain name server. A client would think that he is communicating or receiving the information from a genuine web site, though it would not be the case actually.

5. Denial of services (DOS) attacks

In this type of attack, the attacker gains access to the network and then send invalid data to network services or applications. These services or applications consequently become unable to perform their normal tasks or functions. Hence, sending a flood of data to a particular service or computer can cause it to overload or shutdown. This attack is specially used to take down websites on the internet, when repeated requests for web pages are deliberately initiated so as to choke down a web server.

Question No: 17 (Marks: 1)

What is the function of TCP protocol at TCP/IP stack? Give precise answer.

The **User Datagram Protocol (UDP)** is one of the core members of the Internet Protocol Suite, the set of network protocols used for the Internet. With UDP, computer applications can send messages, in this case referred to as datagram's, to other hosts on an Internet Protocol (IP) network without requiring prior communications to set up special transmission channels or data paths.

UDP is a simpler message-based connectionless protocol. Connectionless protocols do not set up a dedicated end-to-end connection.

Question No: 18 __ (Marks: 1)

What do you say that Cookies can also be used to maintain state between two broken sessions? Justify with solid argument

Cookies can also be used to maintain state between two broken sessions. Thus, it is possible that some information of the previous session is stored in the cookie file, and this information is available (through cookie) to the server side on starting a subsequent session.

Question No: 19 __ (Marks: 2)

What is bandwidth of a network?

A Bridge is a combination of hardware and software. It has its own operating system. It helps to conserve the bandwidth of a network. Bandwidth is the speed of a network. It indicates how many bits can be transmitted across a network in a given time.

Question No: 20 (Marks: 3)

What is the purpose of padding? How many types of padding are: Name of these?

Padding-top – adds space on top of the content

Padding-right – adds space to the right of the content

Padding-bottom – adds space below the content



Padding-left – adds space to the left of the content

Padding – general property that applies to all four padding properties in this order: 'padding-top, padding-right, padding-bottom, padding-left'

Question No: 21 __ (Marks: 5)

Write any four essential features and two extra features which should exist in E-Commerce soft wares.

A catalog display

A Static catalog is a simple list written in HTML. It can provide photos, detailed description and search facility about the availability of items.

Shopping cart capability

It is now replaced by electronic shopping cart/basket that keeps record of the items the customer has selected and allows customers to view the details of the items selected. The customer can add new items or remove items. To order an item, a customer simply clicks at that item option. All details of an item, including its price, product no. and order identifying information are stored automatically in the cart.

Transaction processing

It occurs when a customer clicks at checkout or confirm order button and is subjected to some payment processing mechanism. It represents the most complex part of online sale. Calculation of taxes, shipping costs etc. is important parts of this process.

Web services

Web services are defined as a combination of software tools that allow application software in one organization communicate with other programs/applications over a network by using a specific set of standard protocols. Web services can be used to obtain price and delivery information about goods from different vendors/suppliers, review this information, place the order to the right vendor/supplier and track the order till shipment is received.

Application integration

A program that performs a specific function such as creating invoices/bills or processing payment received from customers is called an application program.

Middleware

Large companies establish connections between their e-commerce software and their existing accounting system by using a type of software called Middleware which is a part of e-commerce software package.

Question No: 17 (Marks: 1)

What is the function of ARP protocol at TCP/IP stack? Give precise answer.

ARP: Address resolution protocol map IP address to MAC hardware address.

Question No: 18 __ (Marks: 1)

Do you agree or not that in In Java Script we are not provided with certain already defined objects which are ready to use? Justify



In Java Script we are provided with certain already defined objects which are ready to use. JavaScript Reserved Keywords are part of the JavaScript language syntax. JavaScript Reserved Keywords has special meanings and used to instruct JavaScript interpreter to perform a specific pre define task. For example Alert is one of the keyword, when we use Alert function in java script, Browser immediately display Alert box.

Question No: 19 (Marks: 2)

What are the three position style properties?

Position Property

Position: absolute – you can place the element box anywhere on the page position:

Relative – position is relative to the normal position of the element

Position: static – position remains unchanged

Question No: 20 (Marks: 3)

Write at least three rules for writing an XML code.

Rules for writing an XML code

-All elements must be properly nested <outer><inner>content</inner></outer>

-All attribute values must be quoted <FRIES SIZE="LARGE">

-All elements with empty content must be identified by ending in />

Question No: 21 (Marks: 5)

What is the difference between an interpreted and a compiled language? Give one example of each.

A compiled language is written and then run through a compiler which checks its syntax and compresses it into a binary executable. Since an interpreted language is not compiled, it must be checked for errors at run-time, which makes it quite a bit slower than a compiled language (like C or Java). Perl is an example of an interpreted language. Remember, though, that just because a language is interpreted doesn't necessarily mean it is not full-featured, or simplistic. Perl can get very complex and very cryptic, very quickly. C, C++ is the examples of complied language.

Question No: 22 __ (Marks: 10)

Consider the following code of an XML document and identify where its standard code rules are being applied. You are required to write the identified tag along with its related rule side by side.

<?xml version="1.0"?>

<PlanetsList>

<Planet Name="Mercury"> <Distance>36 million miles</Distance>

<Moons>None</Moons>

<DayLength> 176 days</DayLength> </Planet>



```

<Planet    Name="Venus">    <Distance>67    million    miles</Distance>
<Moons>None</Moons>
<DayLength>1 17 days</DayLength> </Planet>
<Planet    Name="Earth">    <Distance>93    million    miles</Distance>
<Moons>One</Moons>
<DayLength>24 Hours</DayLength> </Planet>
</PlanetsList> <BR/>

```

First line of the code is a declaration that it is an XML document (version 1). Second and last lines of the code are called root element tags. We enclose other elements within the root element tags. We assign a name to the root element that best describes the purpose of our file. Other elements are called child elements. Thus, planet is a child element of planet list. Further, each property of a planet is the child element of the planet element. So, distance, moons and day length are the child elements of planet element. Name is the attribute of the planet element. Names of child elements can be different between two organizations, which can make the sharing of information difficult. For instance, some may describe the property of a planet as Day and others may use the word Day length for that purpose.

Question No: 17 (Marks: 1)

What does this mean padding-right?

Padding-right – adds space to the right of the content

Question No: 18 (Marks: 1)

Do you agree or not that using Cascading Style Sheets one can apply an image to sections of one's page or even to individual words.

Yes m agree that Using Cascading Style Sheets one can apply a background color or image to sections of one's page or even to individual words.

For more detail: page 47

Question No: 19 (Marks: 2)

Suppose there is an applet at "<http://www.vu.edu.pk/demo.class>". Write the code to include this applet in your web page.

```

<APPLET CODE= "http://www.vu.edu.pk/demo.class" width=100 height=200>
DEMO </APPLET>

```

Question No: 20 (Marks: 3)



What do you think, is there any difference between Static and Dynamic IP Address assignment? Discuss.

A static IP address is an address that does not change over time unless changed manually. It is used when you need the IP address or network location to remain the same consistently. A good example of this is for a web server. If you go to www.google.com you are really going to the IP address of 66.102.7.99. If this were to change suddenly you would not be able to get to Google unless you knew the new IP address or until Google updated their DNS records.

A Dynamic IP address is an address that keeps on changing. It is mostly used when having a consistent IP address is not necessary. An example of this would be the IP address your ISP (internet service provider) assigns you when you log on to the internet. You must have an IP address to surf the web but once you disconnect you lose that address. Then the next time you sign on you are assigned a new one. This is done using DHCP.

Question No: 21 (Marks: 5)

What is cryptography? What techniques are used for cryptography?

Cryptography is a collection of mathematical techniques used to ensure confidentiality of information. The process of scrambling a message with the help of a key is called Encryption. The process of unscrambling a message using an appropriate key is called decryption

Techniques used for cryptography

Substitution In substitution we replace each letter in the message with another to make the message non-understandable. For example, each letter "a" in the message can be replaced with letter "d" and letter "b" with letter "e" and so on.

Transposition It is based on scrambling the characters in a message. A transposition system may first write a message into a table row by row then the message can be read and rewritten column by column to make it scrambled

Question No: 22 (Marks: 10)

Analyze the following given code showing some Cascading Style Sheets tags and explain its lines one by one.

```
<html><head>
```

```
<style type="text/css">
```

```
h1 {text-decoration: overline}
```

```
h2 {text-decoration: line-through}
```

```
h3 {text-decoration: underline}
```

```
a {text-decoration: none}

</style></head>

<body><h1>This is header 1</h1>

<h2>This is header 2</h2>

<h3>This is header 3</h3>

<p><a href="http://www.w3schools.com/default.asp">This is a link</a></p>

</body></html>
```

h1 {text-decoration: overline} Defines a line above the text

h2 {text-decoration: line-through} Defines a line through the text

h3 {text-decoration: underline} Defines a line below the text

a {text-decoration: none} Defines a normal text

Question No: 17 __ (Marks: 1)

Shortly Explain RIP?

Routing information protocol:

Routing protocol used by routers to determine the best path for packets on the network.

Question No: 18 __ (Marks: 1)

In HTML , for displaying information systematically which tag do we usually use ?

Tables are used largely for page layout as well as for displaying information systematically. Tables use the basic tag <Table> and </Table>. All other table tags fit between these two tags. <TR> (table row tag) and </TR> (corresponding end tag) are used to create/add a row. <TD> (table data tag) and </TD> (corresponding end tag) are used to divide a row into number of columns to create cells. <TH> (table header tag) and </TH> (corresponding end tag) labels each column as a heading. To provide the caption of a table <caption> and </caption> can be used.

Question No: 19 (Marks: 2)

How Certification Authority (CA) issues the certificates? Describe briefly

Question No: 20 __ (Marks: 3)

What is MAC address? How does it help server to identify the client?

Question No: 21 (Marks: 5)

How can we classify E-Commerce by the nature of its transactions? Give at least one real life example for each of its classification.

Question No: 22 (Marks: 10)

Analyze the following given code showing some Cascading Style Sheets tags and explain its lines one by one.

```
<html>
<head>
<style type="text/css">
h1 {text-decoration: overline}
h2 {text-decoration: line-through}
h3 {text-decoration: underline}
a {text-decoration: none}
</style>
</head>
</html>
<body>
<h1>This is header 1</h1>
<h2>This is header 2</h2>
<h3>This is header 3</h3>
<p><a href="http://www.w3schools.com/default.asp">This is a link</a></p>
</body>
```

h1 {text-decoration: overline} Defines a line above the text

h2 {text-decoration: line-through} Defines a line through the text

h3 {text-decoration: underline} Defines a line below the text

a {text-decoration: none} Defines a normal text

Question No: 17 __ (Marks: 1)

What does this mean document.write("str.charAt(3) : "+str.charAt(3) + "<P>"); ?

CharAt() is the function/method of string object.

charAt(3) would provide the value of the element at the index three.

document.write("str.charAt(3) : "+str.charAt(3) + "<P>") Mean that character string at index number three should return to me

Question No: 18 __ (Marks: 1)

What does this term mean "Non-repudiation"?

Non-repudiation means that the execution of the digital signatures cannot be denied by the one who is alleged to be the executor of those signatures.

Question No: 19 __ (Marks: 2)

What are servlets?

Servlets are very fast Java applications on the server side which are available in an active form in the memory of the server. They use JDBC to connect to the databases.

A **Servlet** is a Java class in Java EE that conforms to the **Java Servlet API**, a protocol by which a Java class may respond to HTTP requests. They are not tied to a specific client-server protocol, but are most often used with this protocol.

Question No: 20 (Marks: 3)

What are the three methods for using style sheets with a web page?

Internal Stylesheets

This method places the CSS code itself inside the tags of your page. With this method each page is separate; therefore, the code must be placed inside each page in order for the styles to work their magic on every page of your website.

Example:

```
<!--p { font-family: Verdana, Arial, Helvetica, sans-serif; font-size: 12px; color: #000000; h1 { font-family: Arial, Helvetica, sans-serif; font-size: 14px; color: #000099; }
```

Method # 2: Linking to an external style sheet

A separate file is created with the extension '.css', containing your style definitions (see mystyles.css below). Use <Link> tag inside a web page header to apply these style definitions.

mystyles.css:

```
h1 {color: red}
```

```
P {font-size: 34pt}
```

```
Example <HTML> <head> <Link Rel="stylesheet" Type="text/css" href="mystyles.css"> </head> <body><h1>this heading will appear red</h1> <p> this line has a 34pt font</p></body> </html>
```

Method # 3: Inline styles

It is used where a particular tag is to be given a different style. We use 'style' as an attribute in a tag and give properties/values in a format as shown in the example below:

```
<HTML><body><h1 style="font-size: 40pt">this heading has font-size 40-point</h1> </body></HTML>
```

Question No: 21 __ (Marks: 5)

Identify protocol stack used in each layer of OSI model?



Answer:

Handouts page 7 fig.3

Question No: 22 (Marks: 10)

"The role of a Certification Authority is similar to a passport office." Explain it.

You may have noticed that the role of a Certification Authority is analogous or similar to a passport office. The issuance of passport by the passport office attaches credibility that this particular person is entitled to travel. However, the passport is not issued by the office until detailed enquiry/verification about the identity of the person is made. Once a person holds the passport, that confirms that this particular person whose, name, address etc. is appearing on the passport is entitled to travel. Similarly, if a digital certificate is issued by a reputed CA that would confirm to other people that this particular public key certified by the CA belongs to this individual only.

Question No: 17 __ (Marks: 1)

What it stands for (W3C)?

Answer

World Wide Web Consortium

Question No: 18 (Marks: 1)

Do you agree or not that using Cascading Style Sheets one can apply an image to sections of one's page or even to individual words.

Yes I agree that Using Cascading Style Sheets one can apply a background color or image to sections of one's page or even to individual words.

For more detail: Handouts page 47

Question No: 19 (Marks: 2)

Which attribute is used to describe, how many times you want the sound to repeat. Discuss briefly.

Loop="value" - tells the browser how many times to play the file (you can keep the value a number or infinite)

`<bgsound loop="">`

The loop attribute is used to define how many times the sound will repeat itself. Between the quotes specify how many times you want the sound to repeat. Setting loop to "infinite" or "-1" will cause the sound to repeat indefinitely.

Code: `<bgsound loop="5"></bgsound>`

Question No: 20 __ (Marks: 3)

What are cookies? Why these are used in web development?

Cookies are small amounts of data stored by the web browser. They allow you to store particular information about a user and retrieve it every time they visit your pages. Each user has own unique set of cookies.

Cookies are typically used by web servers to perform functions such as tracking your visits to websites, enabling you to log in to sites, and storing your shopping cart. However we don't need fancy web server programming to use cookies. We can use them in JavaScript, too!

Question No: 21 (Marks: 5)

What is the difference between physical address and logical address? Support your answer with at least two examples.

A Physical address is a 48-bit flat address burned into the ROM of the NIC card which is a Layer1 device of the OSI model. This is divided into 24-bit vendor code and 24-bit serial address. This is unique for each system and cannot be changed.

A Logical address is a 32-bit address assigned to each system in a network. This works in Layer-3 of OSI Model. This would be generally the IP address.

Simply we can say that Physical address is MAC address (Media Access Control) which is built-in and Logical Address is the IP which is given manually.

The address generated by CPU is referred to as a logical address. The address seen by the memory unit, i.e, the one loaded into the memory address register of the memory is referred to as physical address

Question No: 22 (Marks: 10)

How can we change a Font style of a text? Write complete tag by using an example by you.

Font tag defines various font properties. 'Size', 'face' and 'color' are its attributes. Attributes can be used simultaneously in a tag as you can see below: internet and ecommerce</ font>

Example code

```
<HTML>
<HEAD></HEAD>
<BODY>
<H1>this is my main title</H1>
<p>here is the first paragraph. This is some <b><i>bold text</i></b></p>
<p><font face="Arial">this text is in the Arial font. </font>
</BODY> </HTML>
```

Question No: 21 (Marks: 2)

What value will be printed after executing the following statement in JavaScript:
`document.write( Math.pow(3, 2) );`

In this function math which is object describe the method of pow and 3, 2 are the arguments of object to method. So if we printed after executing the above statement in JavaScript then resulted value will be 9

Question No: 22 (Marks: 2)

What is a web server?

Web server

Web server is a machine that serves up the web page requests of a client on the internet. It is a combination of hardware and software. Decision as regards the type of a web server is largely dependent on the volume and type of web activities. So, transaction processing sites generating dynamic pages with high traffic volumes would need web server software that is more efficient and can easily be upgraded.

Question No: 23 (Marks: 3)

To calculate the area of a circle the formula πr^2 is used. How would you code it in JavaScript?

```
<html>
<script                                     type="text/javascript">

var                                         radius,area;

radius                                   =                               10;
area                                   =                               Math.PI*Math.pow(radius,2);

document.write("Area of circle with radius 10 is " + area );

</script>

</html>
```

Question No: 24 (Marks: 3)

Name four commonly used web server operating systems.

1. Microsoft Windows NT Server
2. Microsoft Windows 2000 Advanced Server
3. Microsoft.NET Server
4. Linux or Unix based operating systems such as Solaris

Question No: 25 (Marks: 5)

Write the types of firewalls. Only names? Explain these briefly

There are three main types of firewalls detailed as follows:

1. Packet filter firewall

It uses a set of rules to determine whether outgoing or incoming data packets are allowed to pass through the firewall

2. Circuit level firewall

Works on the basis of a set of rules for filtering packets but operates at the Transport layer of the OSI Model so has greater functionality. It can make packets sent from internal network to a destination outside the firewall appear as if they originated at the firewall.

3. Application gateway firewall

It operates at application layer of the OSI Model. It uses strong user authentication to verify identity of a host attempting to connect to the network using application layer protocols such as FTP. In contrast to packet filter firewall, it filters the requests rather than packets entering/leaving the network. It can block any outgoing HTTP or FTP requests

Question No: 26 (Marks: 5)

Write the JavaScript codes which will display the following in a web page using a FOR loop:

The number is 0

The number is 1

The number is 2

The number is 3

The number is 4

The number is 5

Answer:

```
<HTML>
<TITLE>Using the For Statement</TITLE>
<BODY>
<script type="text/javascript">
var no;
for (no=0; no<6 ; no++)
{
document.write("The number is " + no + "<br>");
}
</script></BODY>
</HTML>
```

Question No: 21 (Marks: 2)

What you understand about Denial of services (DOS) attacks?

In this type of attack, the attacker gains access to the network and then send invalid data to network services or applications. These services or applications



consequently become unable to perform their normal tasks or functions. Hence, sending a flood of data to a particular service or computer can cause it to overload or shutdown. This attack is specially used to take down websites on the internet, when repeated requests for web pages are deliberately initiated so as to choke down a web server.

Question No: 22 (Marks: 2)

Discuss the role of (W3C) keeping in view the web documents?

The goal of the W3C is to help the World Wide Web (also known as the Web) evolve by developing common languages and formats that will allow users with different hardware and software to enjoy similar features of the Web. In essence, its objective is to make the Web accessible to all and not dependent on the technology of any single company. This area encompasses defining the Web, Web documents, and the protocols for accessing and disseminating Web documents.

Question No: 23 (Marks: 3)

Differentiate between the HTML and XML. Identify At least three major differences.

The Difference Between XML and HTML

XML is not a replacement for HTML.

XML and HTML were designed with different goals:

XML was designed to transport and store data, with focus on what data is

HTML was designed to display data, with focus on how data looks

HTML is about displaying information, while XML is about carrying information

Question No: 24 (Marks: 3)

What are the different logical operators in JavaScript? Only write their names. Also write the symbols of operators.

Logical operators '&&' for logic AND operator, which returns a Boolean true if both the operands are true

'||' for logical OR operator and it returns a value of true if one or both of the operands is true.

'!' for logical not operator which returns false if its single operand can be converted to true, or if it is a non-Boolean value:

Question No: 25 (Marks: 5)

Write HTML code for the form displaying a drop-down list containing the following entries: Honda, Suzuki, Santro, Corolla.

<Html> <head>

<Title></title>

</head>

```
<Body >
<form action= "http://www.vu.edu.pk" method="post">
<select name="car">
<Option selected> --- Click to Select --- </option>
<Option>Honda </option>
<Option>Santro</option>
<Option>Suzuki</option>
<Option>Corolla</option>
</select> </body> </html>
```

Question No: 26 (Marks: 5)

Write a note on any two methods used to maintain state between HTTP requests.

Answer1:

A cookie is used to store small piece of information on client machine. A cookie contains page-specific information that a Web server sends to a client along with page output. Cookies are used for sending page-specific information because HTTP is a stateless protocol and cannot indicate whether page requests are coming from the same or different clients. You can use cookies to keep track of each individual user who access a Web page across an HTTP connection.

Application state means storing application wide-specific information such as objects and variables. The information in the application state is stored in a key-value pair and is used to maintain data consistency between server round trips and between pages. Application state is created when each browser request is made for a specific URL. After an application state is created, the application-specific information is stored in it. All information stored in the application state is shared among all the pages of the Web application by using the HttpSessionState class.

Answer2:

<http://www.beansoftware.com/ASP.NET-Tutorials/Understanding-State-Management.aspx>

Answer3: Handouts page88**Question No: 21 (Marks: 2)**

How Certification Authority (CA) issues the certificates? Describe briefly

A certification authority is defined to be a trusted public/private body that attests the association of a particular individual with his/her corresponding public

key. A CA signs digital certificates with its private key. There are many CAs working in the field but the pioneering or the most reputed CA is Verisign which is based in America.

Certification authorities work in a hierarchical fashion. There is the CA at the top called root CA (the most reputed CA). It can issue certificates to CAs working below it and those CAs' can further issue certificates to CAs working under them. In this fashion a hierarchy of CAs is developed with each CA confirming the public key of the CA below it through a digital certificate.

Question No: 22 (Marks: 2)

Discuss the role of child element tags while in writing XML code. Names of child elements can be different between two organizations, which can make the sharing of information difficult. A root element is a container for all other elements. While you must have a root to create a viable XML file, it is the child elements that add functionality to the page

Question No: 23 (Marks: 3)

Performance of web servers can be evaluated mainly on the basis of two factors. What are they?

Performance of web servers can be evaluated mainly on the basis of two factors:

Throughput, which is the number of http requests that a particular server hardware and software can handle in the unit time; and

Response time, which is the amount of time a web server requires to process one request. Web server hardware architectures

Question No: 24 (Marks: 3)

Write a function for the addition of two numbers?

```
Function add()  
{  
var var1,var2,var3;  
var1=2;  
var2=4;  
var3= var1+var2;  
document.write(" var1 + var2 = " + c + "<br>");  
}
```

Question No: 25 (Marks: 5)

Write the JavaScript code which will display the following in a web page using a FOR loop:



The number is 0

The number is 1

The number is 2

The number is 3

The number is 4

The number is 5

```
<HTML>
<TITLE>Using the For Statement</TITLE>
<BODY>
<script type="text/JavaScript">
var no;
for(no=0; no<6 ; no++)
{
document.write("The number is " + no + "<br>");
}
</script></BODY>
</HTML>
```

Question No: 26 (Marks: 5)

What is the difference between an interpreted and a compiled language? Give one example of each.

A compiled language is written and then run through a compiler which checks its syntax and compresses it into a binary executable. Since an interpreted language is not compiled, it must be checked for errors at run-time, which makes it quite a bit slower than a compiled language (like C or Java). Perl is an example of an interpreted language. Remember, though, that just because a language is interpreted doesn't necessarily mean it is not full-featured, or simplistic. Perl can get very complex and very cryptic, very quickly. C, C++ is the examples of compiled language.

Question No: 21 (Marks: 2)

We want to create left margin of 10 pixels and right margin of 15 pixels. Write css code to apply both properties.

PG50

```
<HTML><STYLE>P {margin-left: 10px; margin-right: 15px} </STYLE>
<BODY></BODY></HTML>
```

Question No: 22 (Marks: 2)

JDBC and ODBC Stands for?

ODBC (Open database connectivity) and JDBC (Java based database connectivity)

Question No: 23 (Marks: 3)

Write a brief note on ERP? Also write any two ERP software names.

Enterprise Resource Planning (ERP) is a concept that integrates all aspects of a business e.g, accounting, logistics, manufacturing, marketing, planning, project management etc. at a single place. An ERP system such as SAP is expensive. E-commerce sites/software has to fully integrate with ERP software, wherever it is used.

Enterprise Resource Planning is a computer based system which manages internal and external resources of business that includes assets, financial issues, project management, human resource and marketing.

The purpose of this software is to regulate the flow of information among all business functions. It builds a centralized database and it has the common platform. ERP System creates unique and enterprise system environment.

The names of ERP Software

SAP

SAGE

Microsoft Dynamics

Question No: 24 (Marks: 3)

Create an instance of date and write the methods to get date and minutes?

getHours() Returns the current number of hours into the day: (0-23)

getMinutes() Returns the current number of minutes into the hour: (0-59)

getSeconds() Returns the current number of seconds into the minute: (0-59)

getDay() Returns the number of days into the week: (0-6)

getMonth() Returns the number of months into the year: (0-11)

getYear() Returns the number of years into the century: (0-99)

curdate.getDate() Here curdate is an instant

Question No: 25 (Marks: 5)

Write any four essential features and two extra features which should exist in E-Commerce soft wares.

A catalog display

A Static catalog is a simple list written in HTML. It can provide photos, detailed description and search facility about the availability of items.

Shopping cart capability

It is now replaced by electronic shopping cart/basket that keeps record of the items the customer has selected and allows customers to view the details of the items selected. The customer can add new items or remove items. To order an

item, a customer simply clicks at that item option. All details of an item, including its price, product no. and order identifying information are stored automatically in the cart.

Transaction processing

It occurs when a customer clicks at checkout or confirm order button and is subjected to some payment processing mechanism. It represents the most complex part of online sale. Calculation of taxes, shipping costs etc. is important parts of this process.

Web services

Web services are defined as a combination of software tools that allow application software in one organization communicate with other programs/applications over a network by using a specific set of standard protocols. Web services can be used to obtain price and delivery information about goods from different vendors/suppliers, review this information, place the order to the right vendor/supplier and track the order till shipment is received.

Application integration

A program that performs a specific function such as creating invoices/bills or processing payment received from customers is called an application program.

Middleware

Large companies establish connections between their e-commerce software and their existing accounting system by using a type of software called Middleware which is a part of e-commerce software package.

Question No: 26 (Marks: 5)

What are the three logical operators in JavaScript? Write their symbols and their functions.

“++” FOR LOGICAL INCREMENT

“&&” FOR LOGICAL AND

“!” FOR LOGICAL NOT

Logical increment adds one to the previous number that holds a given variable, logical AND returns true when all the given statements are true and logical NOT returns true only when the given statement is not true otherwise it returns false

Question No: 21 (Marks: 2)

What is Term ‘Catalog Display’ means?

Catalog display

A Static catalog is a simple list written in HTML. One has to edit the HTML code to change it. A dynamic catalog stores information about items in a database usually on a different computer accessible by the web server. It can provide photos, detailed description and search facility about the availability of items.

Question No: 22 (Marks: 2)

What is meant by a constructor?

Constructor is defined as the initializing function used to create instance/copy of an object. It is after the name of the object whose constructor it is.

For more detail:

Answer: <http://leepoint.net/notes-java/oop/constructors/constructor.html>

Question No: 23 (Marks: 3)

What is meant by 'Shopping Cart' in context of e commerce?

Shopping cart

In early days form based shopping cart was used wherein the user had to remember and type certain information. It is now replaced by electronic shopping cart/basket that keeps record of the items the customer has selected and allows customers to view the details of the items selected. The customer can add new items or remove items. To order an item, a customer simply clicks at that item option. All details of an item, including its price, product no. and order identifying information are stored automatically in the cart.

Question No: 24 (Marks: 3)

We have an array named myArray where this sentence is stored. "Welcome to Virtual University!" If we apply str.split (' ') method, What will be the output of str.split("")myArray(4) and str.split("")myArray(2) respectively.

Answer:

Str.Split split the string based on the blank spaces.

str.split ("") myArray (4) = blank

str.split ("") myArray (2) = virtual

Question No: 25 (Marks: 5)

Describe the importance of Cryptography in internet data transfer security in E-Commerce?

It is a collection of mathematical techniques used to ensure confidentiality of information. The process of scrambling a message with the help of a key is called Encryption.

It provides the Transaction Integrity. Transaction detail of both the parties can be verified which help to built the confidence between business communities. It provide Privacy like it provide controls to prevent third parties from getting access to the information within the transaction

Without cryptography, it is doubtful that banks, businesses and individuals would feel safe doing business online. Cryptography provides the confidence to the customer that their data is safe and secure. Otherwise due to lack of confidence and doubt e-commerce business can not flourish.

Cryptography, then, not only protects data from theft or alteration, but can also be used for user authentication. There are, in general, three types of cryptographic schemes typically used to accomplish these goals: secret key (or symmetric) cryptography, public-key (or asymmetric) cryptography, and hash functions

Question No: 26 (Marks: 5)

Write a list of comparison operators in JavaScript with symbols.

Comparison operators

1. '==' for Equal
2. '!=' for not equal
3. '<' for Less than
4. '<=' for less than equal
5. '>' for Greater than
6. '>=' for Greater than equal

Question No: 21 (Marks: 2)

What is a web server?

A **web server** is a computer programs that delivers (serves) content, such as web pages, using the Hypertext Transfer Protocol (HTTP), over the World Wide Web.

Any computer can be turned into a Web server by installing server software and connecting the machine to the internet. Every web server has a unique address so that other computers connected to the internet know where to find it on the vast network.

Question No: 22 (Marks: 2)

What does 'parseFloat(string)' And NaN mean?

1. NaN - 'not a number' : Special numeric value NaN (Not a Number) which is used when another type is converted to a number but that conversion does not result in a meaningful number. NaN is always false.

The parse Float function accepts a string argument and returns a floating point number resulting from parsing that string. 'parseFloat(string)' that parses the string as a floating point number, is the example of a function/method of Global Object.

2. The `parseFloat()` function parses a string and returns a floating point number. This function determines if the first character in the specified string is a number. If it is, it parses the string until it reaches the end of the number, and returns the number as a number, not as a string.

The NaN property represents "Not-a-Number" value. This property indicates that a value is not a legal number.

Question No: 23 (Marks: 3)

Write at least three rules for writing an XML code.

Rules for writing an XML code

- All elements must be properly nested
`<outer><inner>content</inner></outer>`
- All attribute values must be quoted `<FRIES SIZE="LARGE">`
- All elements with empty content must be identified by ending in `/>`

Question No: 24 (Marks: 3)

How 'Man in The middle Attacks' operates?

Man in the middle attacks

In it the attacker is able to monitor, capture and control data between sending and receiving machines. He may apply IP address spoofing technique to divert the packets to its machine, then modify the packets and resend the misleading information to the actual client. Another form of man-in-the-middle attack is where the hacker is able to substitute the IP address of a genuine web site with the IP address of his own web site due to some security hole in the software that runs on a domain name server. A client would think that he is communicating or receiving the information from a genuine web site, though it would not be the case actually.

Question No: 25 (Marks: 5)

Write the code for a web page displaying a button named "Testing". After clicking that button an alert should be displayed saying "The test is successful".

```
<html>
<head><script type="text/javascript">
function sndAlert()
{window.alert("The test is successful");}
</script></head>
<body>
<FORM NAME="form1">
<INPUT TYPE="Button" name="Testing" VALUE="Testing"
onclick="sndAlert()">
</FORM>
</body>
```

</html>

Question No: 26 (Marks: 5)

Differentiate between front end and back end.

Most ecommerce sites consist of a front end and a back end. Front end consists of that information which is visible to the user and he can interact with the same through some interface. The back end comprises that information which resides on the server side and is used to create certain dynamic results for the user.

Question No: 21 (Marks: 2)

What does PPTP stand for?

Answer:

PPTP stands for Point to Point Tunneling Protocol

Question No: 22 (Marks: 2)

What are the keywords to define a variable and a function in JavaScript?

Answer:

A variable in JavaScript can be defined using the keyword 'var' and a function by the Keyword 'function'.

Question No: 23 (Marks: 3)

Identify the object, property and method in the following:

Math.sin(45)

Math.PI

Answer:

Math is the object in above case

PI is the property of Math object

Sin () is the method of Math object

Question No: 24 (Marks: 3)

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